

SPIRITWOOD WEST, SK, Canada

WMO: 711330

Lat: 53.3661N

Lon: 107.5564W

Elev: 584

StdP: 94.50

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.5	(c) -31.1	(d) -38.1	(e) 0.1	(f) -34.3	(g) -34.5	(h) 0.2	(i) -30.9	(j) 11.4	(k) -6.6	(l) 10.1	(m) -9.0	(n) 1.8	(o) 160	(p) 0.704

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.3	(c) 27.9	(d) 18.4	(e) 26.1	(f) 17.6	(g) 24.5	(h) 16.8	(i) 20.6	(j) 25.4	(k) 19.2	(l) 23.9	(m) 18.0	(n) 22.7	(o) 4.5	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 18.9	(b) 14.7	(c) 23.4	(d) 17.5	(e) 13.5	(f) 21.5	(g) 16.3	(h) 12.4	(i) 20.3	(j) 62.3	(k) 25.4	(l) 57.4	(m) 24.0	(n) 53.4	(o) 22.6	(p) 25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.0	8.9	7.9	-38.8	31.4	3.3	2.5	-41.2	33.2	-43.1	34.7	-45.0	36.1	-47.3	37.9
			-38.8	22.8	3.2	1.8	-41.1	24.0	-43.0	25.0	-44.7	26.0	-47.0	27.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.5	-15.1	-13.3	-7.0	2.4	9.5	14.2	16.7	15.3	10.5	3.1	-6.2	-13.2		
	DBStd	13.01	9.26	7.99	8.01	5.80	4.59	3.32	2.81	3.26	4.32	4.93	6.92	8.32		
	HDD10.0	3733	779	653	526	233	65	5	0	2	46	218	484	721		
	HDD18.3	6177	1038	887	784	477	275	128	66	102	237	471	734	979		
	CDD10.0	628	0	0	0	6	50	132	207	168	59	6	0	0		
	CDD18.3	31	0	0	0	0	1	5	14	9	1	0	0	0		
	CDH23.3	607	0	0	0	3	44	96	219	186	57	2	0	0		
	CDH26.7	115	0	0	0	1	8	18	37	41	11	0	0	0		
(14) Wind	WSAvg	3.7	3.6	3.6	3.8	4.1	4.1	3.7	3.3	3.2	3.6	3.8	3.7	3.5		
(15) Precipitation	PrecAvg	456	20	14	17	30	40	83	87	61	40	26	20	18		
	PrecMax	629	36	30	32	81	104	139	139	120	93	63	48	39		
	PrecMin	321	7	4	3	4	8	37	37	13	11	5	4	8		
	PrecStd	65	8	6	8	18	22	28	26	25	22	13	11	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.3	4.9	10.7	21.7	27.6	28.7	30.1	30.8	28.2	21.5	11.2	5.9		
		MCWB	2.3	2.0	5.9	11.1	14.7	17.9	20.6	19.5	17.0	13.0	6.7	2.5		
	2%	DB	3.3	2.5	7.2	18.1	24.0	25.8	27.4	27.5	24.8	16.7	7.5	3.3		
		MCWB	1.2	0.2	3.6	9.2	12.8	16.7	19.7	18.6	15.2	10.1	4.2	1.2		
	5%	DB	1.0	0.3	5.2	15.0	21.6	23.8	25.8	25.3	21.9	13.7	4.8	1.0		
		MCWB	-0.6	-1.2	2.4	7.8	11.9	15.6	18.7	17.7	13.9	8.1	2.4	-0.7		
	10%	DB	-1.5	-1.6	3.3	11.9	19.3	22.0	24.1	23.4	19.1	11.0	2.5	-1.2		
		MCWB	-2.7	-2.7	1.2	6.4	11.0	14.8	17.8	16.9	12.6	7.1	0.9	-2.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	2.3	6.2	12.0	16.4	20.5	23.0	21.5	18.4	13.5	6.8	3.1		
		MCDB	4.9	4.5	10.2	20.4	22.8	25.9	27.3	26.8	24.2	20.3	10.9	5.1		
	2%	WB	1.4	0.4	3.9	10.1	14.2	18.4	21.3	19.9	16.2	10.7	4.7	1.3		
		MCDB	3.1	2.1	6.8	16.7	21.5	23.3	25.6	25.2	22.7	15.1	7.2	3.2		
	5%	WB	-0.5	-1.1	2.6	8.4	13.0	17.1	19.8	18.6	14.6	8.8	2.7	-0.6		
		MCDB	1.0	0.2	4.8	13.9	19.9	21.6	24.1	23.8	20.5	13.0	4.6	1.0		
	10%	WB	-2.7	-2.7	1.3	6.7	11.6	15.8	18.6	17.4	13.2	7.2	1.0	-2.5		
		MCDB	-1.5	-1.5	3.2	11.4	18.0	20.4	22.8	22.3	18.2	10.4	2.3	-1.2		
(35) Mean Daily Temperature Range	MDBR		10.1	11.2	10.8	11.2	13.3	11.9	12.3	12.9	12.7	10.4	8.6	9.6		
	5% DB	MCDBR	10.6	12.0	11.0	16.2	17.5	15.7	15.5	16.9	18.3	15.7	11.2	10.9		
		MCWBR	9.1	10.1	8.0	9.1	8.7	8.0	8.8	9.1	9.8	9.6	8.2	9.2		
	5% WB	MCDBR	10.6	11.8	10.6	15.2	15.4	13.2	14.2	15.1	16.4	14.7	10.6	10.5		
		MCWBR	9.2	10.1	8.0	9.1	8.4	7.9	9.1	9.1	9.5	9.3	8.1	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.233	0.256	0.277	0.327	0.345	0.363	0.363	0.359	0.308	0.276	0.244	0.224		
	taud		2.333	2.305	2.369	2.390	2.388	2.371	2.376	2.378	2.526	2.578	2.466	2.345		
	Ebn at Noon		780	865	920	907	903	887	880	862	877	845	772	726		
	Edh at Noon		55	79	93	105	111	114	112	105	80	61	48	44		
(44) All-Sky Solar Radiation	RadAvg		0.97	2.04	3.58	4.76	5.74	5.79	5.95	4.90	3.44	2.00	0.91	0.69		
	RadStd		0.09	0.11	0.24	0.45	0.48	0.45	0.39	0.40	0.39	0.33	0.14	0.08		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page